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18 November 1976
SUPERSEDING
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30 June 1964

MILITARY SPECIFICATION

SURVEYING INSTRUMENT, DISTANCE-MEASURING, ELECTRONIC MICROWAVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, Surveying Instrument, Distance Measuring, Electronic Microwave (SEDME). SEDME units shall be of solid-state construction and are complete with accessories for unsupported operation.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-C-440	- Clip, Electrical, General Specification for.
QQ-S-781	- Strapping, Steel and Seals.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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ZZ-R-765	- Rubber, Silicone: Low- and High-Temperature and Tear Resistant.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-C-4150	- Cases, Transit and Storage, Waterproof And Water-Vaporproof.
MIL-E-4158	- Electronic Equipment Ground; General Requirements for.
MIL-E-5272	- Environmental Testing, Aeronautical and Associatied Equipment, General Specification for.
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-H-12576	- Headset-Microphones H-91 ( )/U and H-92 ( ) /U and Handset-Headset H-144 ( )/U.
MIL-T-14091	- Tripod, Surveying: Extension-Leg-Type.
MIL-T-14132	- Theodolites, Surveying, Direction: Second Order.
MIL-L-25467	- Lighting, Integral, Aircraft Instrument, General Specification for.
MIL-C-26482	- Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting.

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MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, for Packaging.
MIL-P-55110	- Printed Wiring Boards.
MIL-B-55118	- Batteries, Storage, (Cells), Vented, Nickel-Cadmium.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-681	- Identification Coding and Application of Hookup and Lead Wire.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as dirrected by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated,

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the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

COUNTY OF LOS ANGELES - AIR POLUTION CONTROL DISTRICT.

Rule 442 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

TB MED 270/AFM 161-7

Control of Hazards to Health From Microwave Radiation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Description. The SEDME shall consist of an antenna; electronics package (control unit); transport case; psychrometer; altimeter-barometer; meteorological correction chart; 12 volt NiCad battery and charger; external power cables, one not less than 8.0 meters (26.2 feet) in length and when the NiCad battery is an external power supply (see 3.9.1.1) a cable 3 meters (10 feet) long shall also be necessary; spare fuses; tribrach; repair kit; backpacks (2); tripod; adapter unit which will enable the antenna to be attached to the control unit and a tribrach and allow it to be adjusted vertically and horizontally for maximum signal return but still keep the instrument's zero point plumbed over the marker; two antenna connecting cables of which one must be at least 37 meters (121.4) feet long and the other cable shall not be longer than 1 meter (3 feet) in length; accessories and accessory case; when specified (see 6.2) a winterization kit shall be included.

(a) SEDME shall measure distances to the required accuracy (see 3.6.3) by transmitting a frequency modulated signal to and receive a frequency modulated signal from another SEDME, compute the slope distance, by measuring the phase shift between the two signals and then display the corrected slope distance in meters.

(b) SEDME shall be capable of measuring in the following configurations:

(1) The antenna indirectly mounted on the control unit via the adapter unit.
(2) The antenna with the adapter unit indirectly mounted on a tripod with the adapter unit indirectly mounted on a tripod (MIL-T-14091) via a tribrach (MIL-T-14132).
(3) The antenna mounted on a mast.

When the SEDME is in the last two configurations, an antenna separation from the control unit of at least 37 meters (121.4 feet) shall be possible.

(c) The range of the SEDME shall be from 200 meters to 50 kilometers. The SEDME shall have a built-in, single channel communication system for voice communication during the measurement processes. SEDME shall measure distances under the environmental conditions specified herein during the night and day. The control unit shall be indirectly mountable on a tripod (MIL-T-14091) via a tribrach (MIL-T-14132).

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3.2 First article (preproduction model). The contractor shall furnish two or more SEDME as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce SEDME that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more SEDME for inspection as specified in 4.4.

3.4 Materials. Materials shall be free from all defects and imperfections that may affect the operation or life of the instrument. Materials used shall be both lightweight and corrosion resistant such as alluminum alloys, magnesium alloys, copper alloys, alloy steel containing a minimum of 11 percent chromium, or molded plastic reinforced with glass fibers. Where the use of corrosion-resistant materials is not practicable, the materials shall be treated to resist corrosion. Contact between dissimilar metals, as defined in MIL-STD-889 that will cause deterioration of parts by galvanic corrosion shall be avoided, or dissimilar metals shall be separated by interposing a material compatible to each. Fungus-inert parts and materials shall be used wherever possible.

3.5 Environmental requirements.

3.5.1 Operational temperature. The SEDME shall operate with no external assistance, without damage from the main and alternate 12-volt power source, with the accuracy specified herein when in thermal equilibrium in air at any temperature from +52° C (+125° F) to -18° C (0° F). The SEDME shall operate without damage and with the accuracy specified herein from -18° C (0° F) to -43° C (-45° F) using an external winterization kit.

3.5.2 Storage temperature. The SEDME shall not be damaged by storage under the following conditions:

Air temperature - +71° C (+160° F)	48 hours
Low temperature - -57° C (- 70° F)	24 hours

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3.5.3 Solar radiation. The SEDME shall operate, without damage, with the accuracy specified herein when subjected to solar radiation of 1136 watts per square meter (w/m^2) [360 British thermal units per square foot per hour (Btu/sq ft/hr)] for not less than 4 hours in an environmental air temperature of +52° C (+125° F), +0° C, -3° C.

3.5.4 Rain. The SEDME shall not be damaged nor show evidence of water penetration and shall operate within the accuracy specified herein, before and after exposure to windblown [65 Kmph (40 mph) ± 8 Kmph (5 mph)] rainfall of 10 cm (4 in.) ±2.5 cm (1 in.) per hour for a period of not less than 2 hours.

3.5.5 Humidity. The SEDME shall not be damaged and shall operate within the specified accuracy when subjected to relative humidities of at least 95 percent at air temperatures varying between +20° C (+68° F) ± 1° C (2° F) and +52° C (+125° F) ± 1° C (2° F).

3.5.6 Sand and dust. When the SEDME transport and accessory cases are subjected to blowing sand and dust, the SEDME shall not show sand and dust penetration into the electronics package. The accessory case and the transport case shall not show sand or dust penetration.

3.5.7 Vibration. The SEDME shall not be damaged by vibration nor lose accuracy specified herein when subjected to the vibration test specified in 4.6.2.12.

3.5.8 Shock. The SEDME shall not be damaged nor lose accuracy specified herein when subjected to the shock test specified in 4.6.2.13.

3.6 Performance.

3.6.1 Precalibration. Each instrument shall be calibrated by the contractor. If the electrical center and the plumb bob center are different, the value of this distance shall be displayed on the instrument identification plate. A means shall be provided for adjusting the instrument to this value without the use of special tools.

3.6.2 Compatibility. Each instrument shall be capable of acting as measurer and responder with any other instrument built to the "C" revision of MIL-S-52156, without any special calibration.

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3.6.3 Accuracy and range. When used in a system of two or more instruments, each instrument shall measure distances from 200 meters to 50 kilometers. Each instrument shall have the following accuracies for the respective range:

200 M to 5KM - 1 part in 7,000
5KM to 15KM - 1 part in 25,000
15KM to 50KM - 1 part in 35,000

3.6.4 Warm-up time. SEDME shall be ready for a measurement within the following time limits for the respective temperature ranges:

-18° C (0° F) to +52° C (+125° F) - 20 minutes
-43° C (-45° F) to < -18° C (0° F) - 30 minutes

A winterization kit may be used for the temperature range of -43° F to < -18° C.

3.6.5 Distance measurement time. After setup and warmup of the SEDME, the time required for distance measurement in ambient temperatures from -43° C to +52° C shall be not more than 7 minutes.

3.6.6 Distance measurement operation. The tuning and switching of the carrier and modulating frequencies, phase shift measurement and total corrected distance measurement shall be done automatically using a microprocessor.

3.7 Physical characteristics.

3.7.1 Portability. The instrument shall be furnished in man-portable packages.

3.7.1.1 Weights. The instrument packages shall weigh not more than the following:

Item	Maximum Weight Kilograms	(Pounds)
Antenna cables (short and long)	6.4	(14.0)
Accessory case	5.7	(12.5)
Altimeter-Barometer	.9	(2.0)
Backpacks	3.0	(6.5)
Cable for alternate power source	2.7	(6.0)
Cable for main power source	.4	(1.0)
Control unit, antenna and adapter unit without main power source	18.2	(40.0)
Main power source	4.5	(10.0)
Psychrometer	2.7	(6.0)
Transport case	30.2	(66.5)

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3.7.1.2 Volume and dimensions.

(a) The volume of the control unit shall not be more than .04 cubic meters (1.4 cubic feet). The length and width shall not be more than 400 mm (15.8 inches) and the depth shall not be more than 250 mm (9.8 inches).
(b) The volume of the antenna unit shall not be more than .018 cubic meters (.64 cubic feet). The diameter shall not be more than 300 mm (11.8 inches) and the height shall not be more than 250 mm (9.8 inches).

3.7.2 Controls. Panel dials, knobs, and switches shall be functionally grouped and in accordance with MIL-E-4158. Dials, knobs, and switches shall be protected. The panel layout shall be submitted to the contracting officer for approval of control location and operation in regard to human factors engineering.

3.7.3 Display unit. The display unit shall provide a digital readout of the corrected slope distance in meters with a resolution of 1 centimeter. The numerical characters shall not be less than 6.35 mm (.25 inch) high.

3.7.4 Receptacles. SEDME receptacles shall conform to MIL-C-26482. One receptacle shall be for the external power supplies (main power supply, if applicable and the alternate power supplies). One receptacle shall be for the headset microphone.

3.8 Electronics package.

3.8.1 Modules. The electronics package shall be designed and arranged in modular units and shall have a positive guide arrangement to facilitate module removal and replacement.

3.8.2 Captive hardware. Captive hardware shall be used on all screws that can be loosened from the exterior of the electronics package.

3.8.3 Component boards. The components shall be mounted on printed wiring boards in accordance with MIL-P-55110. The plug-in component boards shall be equipped with a coded keying arrangement.

3.8.4 Panel marking. Markings on the control panel shall identify each control, meter, and connector by its functional name, symbol, or operating instruction reference.

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3.8.5 Panel illumination. All operating indicators and dial illumination shall be adjustable and controlled by a front panel knob. The color shall be as specified in MIL-L-25467. All lamps except for meters shall be replaceable from the outside of the electronics package.

3.8.6 Electronic components and parts. Electronic components and parts shall conform to MIL-E-4158. No hand-selected components shall be used.

3.8.7 Color coding. Hookup and chassis wiring shall be color coded in accordance with MIL-STD-681.

3.8.8 Test meter. A built-in test meter in accordance with MIL-M-10304 shall be provided for monitoring electrical circuits, such as crystal current, automatic volume control (avc) voltage, dc operating voltages, and other circuits as necessary to insure the instrument is operating normally. When used in conjunction with the maintenance manual, the test meter shall give an indication as to which module or subassembly is not functioning properly. The test meter shall have internal illumination in accordance with MIL-L-25467.

3.9 Power. SEDME shall operate on not more than 60 watts with the power sources listed in 3.9.1 and it shall operate in the voltage range of 11.0 to 34.0 volts D.C. SEDME shall contain built-in protection against reverse polarity from the power source.

3.9.1 Power sources.

3.9.1.1 Main power source. The battery shall be a nickle-cadmium type conforming to MIL-B-55118 with a combination of cells to give 12-volt operation. The battery shall either be an internal or external power source (see 3.9.1.1.1 and 3.9.1.1.2). The battery shall be rechargeable via a 110/220-volt, 60 Hz charger supplied with the system and shall provide power sufficient to operate for 2 hours. A spare battery(s) pack may be used to accomplish the stated mission if it is integrally mounted in the field-portable unit and can be exchanged within 30 seconds. The battery(s) shall be spillproof and shall have no detectable liquid or gas regardless of operation or storage orientation and shall not be damaged when subjected to shock testing as specified in 3.5.8. The battery shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

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3.9.1.1.1 Internal power source. When the main power source is an internal battery, a metal tray complete with quick-plug-in circuitry connector and positive guide arrangement and removable from the front panel shall be provided to hold the power units. The front plate of the tray assembly shall be gasketed and held in place with one-fourth-turn fasteners. The inside surface of the tray and the power unit compartment shall be coated with a fungus-proof protective material in accordance with MIL-STD-202 and MIL-E-5272.

3.9.1.1.2 External power source. When the main power source is an external battery, it shall be attached to the electronics package via a power cable (see 3.10.1) with quick-disconnect fasteners.

3.9.1.2 Alternate power sources. SEDME shall operate off batteries conforming to MIL-B-11188, 12-volt D.C., Class FE for 4 hours and vehicle batteries mounted in a military truck.

3.10 Power cables. One end of each cable shall terminate in a plug conforming to MIL-G-26482, quick-disconnect, straight or angle, size and arrangements as required, which shall fit the instrument receptacle.

3.10.1 Main power cable. If the main power source is an external battery, it shall have a cable which is at least 3 meters (10 feet) long. The other end of the cable shall also terminate in a plug conforming to MIL-G-26482, quick-disconnect, straight, or angle, size and arrangements as required, which will fit the battery output receptacle.

3.10.2 Alternate power cable. The alternate power cable shall be at least 8.0 meters (26.2 feet) long and shall terminate in battery clips conforming to W-C-440, Type PC, with insulators.

3.10.3 Wire and cable marking. Mechanically separable connections shall be indelibly marked with characters of a size and color to insure legibility. Cables shall be identified at the break-out of each branch and at each termination. Wire within cables shall be individually marked at each termination.

3.11 Headset-microphone. A headset-microphone conforming to MIL-H-12576, H-91/U, shall be furnished with each instrument.

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3.12 Cases.

3.12.1 Control unit case. The construction of the control unit case shall be in accordance with MIL-C-4150 except for the drop and immersion provisions. The control unit case shall serve as the field carrying and storage case and shall protect the electronics package against shock, vibration, and environmental conditions as specified herein. The entire case shall be dust and moisture tight. All gaskets shall be in accordance with ZZ-R-765, Class 1a, 50 durometer hardness. A carrying handle shall be fitted to the top of the case. Provision shall be made for mounting the control unit directly and indirectly via tribrach on a tripod specified in 3.14 using a 5/8-11 UNC thread. Threads shall be in accordance with Handbook H28.

3.12.2 Antenna case. The construction of the antenna case shall be in accordance with MIL-C-4150 except for the drop and immersion provisions. The antenna cover shall be permanently attached to the antenna case. The antenna case shall serve as the field carrying and storage case and shall protect the antenna against shock, vibration, and environmental conditions as specified herein. The entire case shall be dust and moisture tight. All gaskets shall be in accordance with ZZ-R-765, Class 1a, 50 durometer hardness. Provisions shall be made for mounting the antenna to the control unit via adapter unit and on a tripod via tribrach with the adapter unit.

3.12.3 Accessory case. An accessory case shall be provided to contain repair parts, tools, headset, battery, power cables, manuals, data sheets, and other equipment necessary for field operation. The case shall conform to MIL-C-4150. A carrying handle shall be placed at the balance point of the case when fully loaded. A loading plan shall be furnished.

3.12.4 Transport case. The instrument and accessory case shall be provided with a transport case conforming to MIL-C-4150. Provision shall be made for storage of the carrying packs (see 3.15) in the transport case. Cushioning material in accordance with MIL-P-26514 shall be provided to protect the contents against shock, vibration, and environmental conditions as specified herein during shipment and storage.

3.13 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished for use with SEDME at temperatures below -18° C (0° F) (see 3.6.4). The winterization kit shall include a cover or

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covers for all controls, switches, dials, and meters and a canvas or cotton duck bag for carrying and storing the winterization kit. The winterization kit shall also operate from the power sources as specified in 3.9.1.2.

3.14 Tripod. A tripod conforming to MIL-T-14091 shall be furnished with wrench, square-point screwdriver, and plumb bob (which shall weigh at least .34 kg (12 oz.)).

3.15 Backpack. A means for transporting SEDME by backpacking shall be provided. The means for backpacking shall be human engineered for ease and comfort of a small- to large-framed, 66 kg (145-lb.) to 102 kg (225 lb.) individual. Canvas, if used, shall conform to CCC-C-419, type and weight to suit the purpose, and water-repellent in accordance with CCC-D-950, Type I, Class B, Inhibitor f, and the color shall be forest green as specified in MIL-T-704. There shall be two backpacks, one for the instrument and one for the accessory case.

3.16 Adapter unit. The adapter unit shall allow the antenna to be mounted on the control unit and on a tribrach (see 3.17). It shall also allow the antenna to be adjusted for maximum return signal in the azimuth and vertical planes, and still keep the instrument's zero point plumbed over the marker.

3.17 Tribrach. A tribrach conforming to MIL-T-14132 with an optical plummet, shall be furnished.

3.18 Psychrometer. The psychrometer shall be a Bendix, Environmental Science Division Psychron Model 566-3 or equivalent.

3.19 Altimeter-barometer. The altimeter-barometer shall be an aneroid barometer and measure the barometric pressure in millemeters of mercury, be temperature compensated and able to convert to altitude (in meters) the barometric pressure. It shall have a resolution of at least 3 millimeters for the barometric pressure and at least 30 meters for the altitude reading. It shall have a barometric range of 630 to 800 mm of mercury. It shall not be more than 8 centimeters (3.2 inches) in diameter and 5 centimeters (2.0 inches) deep. It shall not weigh more than .4 kilogram (1 pound).

3.20 Accessories and tools. The following shall be the minimum furnished with each SEDME as accessories and tools:

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Item	Quantity
Adapter unit	1
Altitude-barometer, mm of Hg graduation	1
Antenna connecting cables	2
Backpack	2
External power cable	1
Instruction manual(s)	1 each
Meteorological correction chart (metric)	2
Psychrometer	1
Repair kits	1
Screw driver (square-point)	1
Spare fuses	5
Tribrach	1
Tripod	1
Wrench	1

3.21 Safety. The SEDME shall not have a power density greater than 10 milliwatts per square centimeter anywhere in the microwave beam or present any uncontrolled hazards to operators, bystanders or maintenance personnel. It shall conform to all the applicable requirements of TBMED-270. A statement from the US Army Environmental Hygiene Agency shall be required (see 6.5).

3.22 Human factors. SEDME shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.23 Reliability. The specified Mean-Time-Between Failure (MTFB) of the SEDME shall be at least 500 hours as tested in 4.6.2.17.

3.24 Maintainability.

3.24.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, and test equipment, as specified (see 6.6 and 6.7), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in Section 4.

3.24.2 Maintenance ratio. SEDME shall have a maintenance ratio of not more than 0.05 when tested as specified in 4.6.2.18.2. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required

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(scheduled and unscheduled) to the total operating time. Man-hours for checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.24.3 Ease of maintenance. SEDME construction shall provide ease of access to batteries and fuses (or reset buttons) such that tools are not required to connect a battery or external power nor to change a fuse. SEDME construction shall provide a self-test display to verify satisfactory operating capability prior to initial distance measurement.

3.25 Electromagnetic interference. SEDME shall be tested in accordance with MIL-STD-462, Methods CEO 3 for conducted emissions and REO 2 for radiated emissions and to the requirements of MIL-STD-461 Figures 14 and 22.

3.26 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.26.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 442, Air Pollution Control District, County of Los Angeles, CA.

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3.27 Identification marking. The SEDME shall be identified in accordance with MIL-STD-130. The marking shall be applied to the SEDME on labels conforming to MIL-P-514, Type I, Style 2 Composition A, Class 2 material. Plates shall be attached by adhesive binder in a location where the plate will be both visible and legible.

3.28 Treatment and painting. The portions of SEDME normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. Unless otherwise specified (see 6.2), the color shall be forest green. Non-permissible galvanic couples, as defined in MIL-T-704, Type A, shall be insulated as defined in MIL-T-704, Type A.

3.29 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.30 Workmanship. All parts, components, and assemblies of the SEDME instrument, tripod, and cases including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All parts shall be finished. There shall be no cracks, sharp bends, dents, or chips in finished parts. There shall be no loose solder or insecure electrical connections. The fit of assemblies and subassemblies shall be made without imposing strain or forcing parts. All threads requiring movement and all sliding and rotatable parts shall move without binding, backlash, or endplay. Clamping actions, when required, shall be positive and, when unclamped, shall not cause drag. Bearing surfaces shall show no scratches or any irregular surface blemishes. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified

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in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified here are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction SEDME shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction SEDME shall be subjected to the tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production SEDME will be selected at random by the Government from the instruments being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production SEDME shall not exclude the remaining instruments from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other

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tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production SEDME to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production SEDME and shall be cause for refusal by the Government to continue acceptance of production instruments until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on SEDME previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all SEDME accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Samples. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 6.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 3 Table II. AQL shall be 1.0 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The SEDME shall be examined as specified herein for the following defects:

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Major

101. Operating frequency band not as specified.
102. Material not as specified.
103. Parts, components, or subassemblies not as specified.
104. Controls not as specified.
105. Illumination not as specified.
106. Read-out not as specified.
107. Batteries not as specified.
108. Accessibility of parts and components for service and maintenance not as specified.
109. Dimensions and weights not as specified.
110. Test points not provided for electrical circuitry.
111. Winterization kit not as specified.
112. Power source not as specified.
113. Backpack portability not as specified.
114. Cases not as specified.
115. Instrument modules not as specified.

Minor

201. Treatment, finishing, or painting not as specified.
202. Workmanship not as specified.
203. Identification or special marking missing or illegible.
204. Technical publications missing (when required).
205. Fuses not as specified.
206. Instrument seals against moisture not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to beginning the tests, the contractor shall prepare the SEDME to operate as specified herein. The test chamber used for the environmental tests shall conform to MIL-STD-810. All tests shall be conducted in sequence indicated in Table II.

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4.6.2.2 Test schedule.

TABLE II. TEST SCHEDULE

Preproduction	Initial Production and Inspection Comparison	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	EXAMINATION	4.6.1	3.1
X	X	X	ACCURACY, RANGE, TIME	4.6.2.3	3.6.3
X	X	X	COMPATABILITY	4.4.6.2.4	3.3.6.2
X	X	-	HIGH TEMPERATURE STORAGE	4.6.2.6.1	3.5.2
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	HIGH TEMPERATURE OPERATION	4.6.2.6.2	3.5.1
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	X	LOW TEMPERATURE	4.6.2.7	3.5.2
X	X	X	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	SOLAR RADIATION	4.6.2.8	3.5.3
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	RAIN	4.6.2.9	3.5.4
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	HUMIDITY	4.6.2.10	3.5.5
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	SAND AND DUST	4.6.2.11	3.5.6
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	X	VIBRATION SEDME IN TRANSPORT CASE	4.6.2.12.1	3.5.7
X	X	X	PERFORMANCE	4.6.2.5	3.6.3

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TABLE II. TEST SCHEDULE (Cont.'d)

1	2	3	4	5	6
X	X	-	VIBRATION SEDME OUT OF TRANSPORT CASE	4.6.2.12.2	3.5.7
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	SHOCK SEDME IN TRANSPORT CASE	4.6.2.13.1	3.5.8
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	SHOCK SEDME OUT OF TRANSPORT CASE	4.6.2.13.2	3.5.8
X	X	-	PERFORMANCE	4.6.2.5	3.6.3
X	X	-	VOLTAGE AND POWER CONSUMPTION	4.6.2.14	3.9
X	X	-	ELECTROMAGNETIC INTERFERENCE	4.6.2.15	3.25
X	X		HUMAN ENGINEERING	4.6.2.16	3.22
-	X	-	RELIABILITY	4.6.2.17	3.23
-	X	-	MAINTAINABILITY	4.6.2.18	3.24
X	X	-	ACCURACY, RANGE, TIME	4.6.2.3	3.6.3

4.6.2.3 Accuracy, range, and time. SEDME shall be tested for accuracy, range, and time in the following setups:

(a) A baseline consisting of two monuments which are at the same altitude (± .5 meter) and 210 (± 10, -0) meters apart; a 10 meter beam graduated in meters with one end of the beam alined vertically over one monument, the other end of the beam alined between both monuments and the slope of the beam not greater than .05 shall be used. Both SEDME shall be positioned vertically over their respective monument or meter graduation, within .5 meter of each other. Measurement shall be made at each meter graduation on the beam. An error between the difference of the consecutive measurements and 1 meter greater than .03 meter or the time to complete the measurements not conforming to 3.6.4 shall constitute failure.

(b) A baseline consisting of a master monument and four station monuments in line; at the same altitude (± .5 meter); with distances ranging from 200 to 900 meters (+ 10, -0 meters) between the master and station monuments; with the station monuments 150 to 300 meters apart and the horizontal distances having an accuracy of 1 part in 35,000, shall be used. A measurement shall be made between the master and each station monument with the corresponding time noted. Accuracy and corresponding time not conforming to 3.6.3 and 3.6.4 shall constitute a failure.

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(c) A baseline consisting of a master monument and remote monument located 45 kilometers (+ 5 kilometers apart) with a horizontal accuracy of 1 part in 35,000 shall be used. SEDME shall be positioned over their respective monuments, a series of three distance measurements taken, and corresponding times noted. Accuracy and corresponding time measurements not conforming to 3.6.3 and 3.6.4 shall constitute failure of the test.

4.6.2.4 Compatibility. SEDME'S shall be tested against one another according to 4.6.2.3.

4.6.2.5 Performance. SEDME shall be inspected for accuracy maintainability prior to and after each mechanical and environmental test specified in Table II. The baseline used for this inspection may be any of the baselines mentioned herein or may be a reference baseline established at the test site. The test shall consist of alining SEDME over the monuments, taking a distance measurement, and noting the time. Accuracy and corresponding time measurements not conforming to 3.6.3, 3.6.4, and 3.6.5 shall constitute failure of the test.

4.6.2.6 High temperature.

4.6.2.6.1 Storage. Procedure I of Method 501.1 in MIL-STD-810 shall be followed. Nonconformance to 3.6.3 thru 3.6.5 shall constitute a failure.

4.6.2.6.2 Operation. Procedure II of Method 501.1 in MIL-STD-810 shall be followed except in step 3 a measurement shall be made. Nonconformance to 3.6.3, 3.6.4, and 3.6.5 shall constitute a failure.

4.6.2.7 Low temperature. Method 502.1 in MIL-STD-810 shall be followed. Nonconformance to 3.6.3 thru 3.6.5 shall constitute a failure.

4.6.2.8 Solar radiation. Procedure I of Method 505.1 in MIL-STD-810 shall be followed except the duration of the test shall be 4 hours and interim measurements shall be made every half-hour during the test for a total of eight measurements. Nonconformance to 3.6.3 thru 3.6.5 shall constitute a failure.

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4.6.2.9 Rain. With the instrument in an operational configuration it shall be subjected to a simulated rain at the rate of 10 ± 2cm per hour for 2 hours consisting of four 30-minute periods. During a 30-minute period, a rainfall of 10 ± 2 cm per hour shall be simulated; a wind source producing a horizontal wind velocity of 65 km per hour shall be turned on for the first 15 minutes after which it shall be turned off; a protective cover may be employed to protect the instrument from the rainfall after the first 5 minutes, for 25 minutes out of each 30-minute period. Each face of the instrument, the accessory case, and the transport case shall be pointed into the wind source during a 30-minute period. After the 2-hour test period is completed, the instrument and the cases shall be checked for evidence of damage to operating parts and penetration of moisture into the electronics package, antenna, and cases, which shall constitute a failure of the test.

4.6.2.10 Humidity. Procedure II of Method 507.1 in MIL-STD-810 shall be followed except that only one 48-hour cycle of Figure 507.1-2 shall be used. Nonconformance to 3.6.3 thru 3.6.5 shall constitute a failure.

4.6.2.11 Sand and dust. Method 510.1 in MIL-STD-810 shall be followed. Nonconformance to 3.5.6 shall constitute a failure.

4.6.2.12 Vibration.

4.6.2.12.1 SEDME in transport case. SEDME and the accessory case along with all accessories shall be packed in the transport case and subjected to Procedure X of Method 514.2 in MIL-STD-810 using test levels of Curve "AX" in Figure 514.2-7 and the time schedule for Curve "AX" in Figure 514.2-7 from Table 514.2-VIII. The SEDME and accessories shall be checked for physical damage. Nonconformance to 3.5.7 shall constitute a failure.

4.6.2.12.2 SEDME out of transportation case. SEDME and the accessory case shall be mounted to the vibration table, out of the transportation case. They shall be vibrated along each axis for a period of 1 hour, through a frequency range of 10 to 55 Hz, with a 0.25 cm double amplitude or maximum acceleration of 1.0g with a sweep time of 15 minutes logarithmically. The sweep time includes an ascending plus a descending sweep of the frequency range. The SEDME and accessories shall be checked for any physical damage, which, if it does exist, shall constitute a failure.

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4.6.2.13 Shock. Both the transport case and the accessory case shall be fully loaded when tested during the shock test in 4.6.2.13.1.

4.6.2.13.1 SEDME in transport case. The transport case and the accessory case shall be subjected to Procedure II of Method 516.2 in MIL-STD-810. At the conclusion of this shock test, SEDME shall be inspected for damage. Evidence of damage (see 6.4), to SEDME componentry, excluding cases, shall constitute failure of the test. SEDME cases shall be examined. Inability of the case(s) to protect SEDME componentry shall constitute failure of the test. Damage to cases shall be allowed provided that it does not prohibit the case from continuing its mission. Examples are as follows:

(a) Superficial hairline cracks;
(b) Deformation of sealing edges less than 0.32 cm end-to-end of case which does not inhibit action, leveling action, or sealing;
(c) Wrinkling of corners which will rebound;
(d) Locking latches distorted but not loose at mountings or allowing case seal to deteriorate and still able to function;
(e) Handles abraded but not deformed; case abrasion but no through-the-wall rupture.

4.6.2.13.2 SEDME out of transport case. SEDME shall be tested out of the transport case according to Procedure V of Method 516.2 in MIL-STD-810. After the test, the SEDME shall be checked for physical damage (see 6.4). If any damage exists, a failure has occurred.

4.6.2.14 Voltage and power consumption.

4.6.2.14.1 Voltage range. SEDME shall be set up on any of the baselines used for the accuracy test, with an adjustable voltage having an output voltmeter as the power source. SEDME shall then be operated with the output voltage set at the following levels: 11, 14, 22, 25, 28, and 34 volts. The accuracy of the measurements at each of the voltages previously defined shall be checked for conformance to 3.6.3. Nonconformance to 3.6.3 shall constitute a failure.

4.6.2.14.2 Power consumption. SEDME-MW shall be set up on a baseline with a voltmeter and ammeter or power meter to monitor the instrument's power consumption during a measurement and warm-up. Nonconformance to 3.9 shall constitute a failure.

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4.6.2.14.3 Battery life.

4.6.2.14.3.1 Main power source. SEDME shall be set up on a baseline used in 4.6.2.3 and measure the distance using the main power source (NiCad battery), until the power is insufficient for operation of the SEDME or the accuracy of the measurements are not in compliance with 3.6.3. The total operating time shall be determined and nonconformance to 3.9.1.1 shall constitute a failure.

4.6.2.14.3.2 Alternate power source. SEDME shall be set up on a baseline used in 4.6.2.3 and measure the distance using the alternate power source (lead acid battery), until the power is insufficient for operation of the SEDME or the accuracy of the measurements are not in compliance with 3.6.3. The total operating time shall be determined and nonconformance to 3.9.1.2 shall constitute a failure.

4.6.2.15 Electromagnetic interference. Methods CE 03 for conducted emissions and RE02 for radiated emissions in MIL-STD-462 and MIL-STD-461, Figures 14 and 22 shall be used. Nonconformance to MIL-STD-462 and MIL-STD-461 shall constitute a failure.

4.6.2.16 Human engineering. SEDME shall be tested for human engineering factors as required in 3.22. During the course of all testing herein, test directors shall scrutinize SEDME to determine hardware compatibility to required operator functions. The following functions shall be noted specifically:

(a) Suitability for backpack.
(b) Operation using arctic mittens.
(c) Adaptability to night operation.
(d) Location and configuration of all controls.
(e) Compatibility of operation with other surveying hardware.
(f) Adaptability to charging battery.
(g) Interchangeability of SEDME componentry with other hardware during operation.
(h) Adaptability to easy "setup" in field operation.
(i) Hardware presents no safety hazard from electronics, or mechanical hazard to operator, maintenance personnel, and others.

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4.6.2.17 Reliability. Using an MTBF of 500 hours, the SEDME shall be tested in accordance with MIL-STD-781, Test Plan VI. The SEDME shall be operated in the following sequence:

(a) Start with SEDME packed for transport.
(b) Unpack and set up over a known baseline.
(c) Perform 30 measurements.
(d) Dismantle SEDME and repack for tranport.
(e) Perform manual transport function for a minimum distance of 5 feet and return to original position.

This sequence will be repeated until a decision can be made from Test Plan VI. The temperature shall be ambient temperature. A failure is defined as any instrument malfunction which the operator cannot, or is not authorized to remedy using the controls, on-equipment tools and parts within 30 minutes and which causes failure to commence operation, cessation of operation, a personnel safety hazard, or degradation of performance below specified levels. Simultaneous related malfunctions are considered as one failure. Malfunctions attributable to operator error shall not be considered as chargeable reliability failures. Operator maintenance shall be limited to minor adjustments and replacement of fuses and batteries.

4.6.2.18 Maintainability.

4.6.2.18.1 Maintenance support evalutaion. Maintenance operation required during initial production testing shall be accomplished to determine conformance to 3.24. Failure of the support elements to maintain the SEDME as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio.

4.6.2.18.2 Maintenance ratio. The maintenance ratio shall be computed during testing specified in 4.6.2.17. Operating time to perform each mission and the active maintenance man-hours required for both scheduled and unscheduled maintenance shall be recorded. Nonconformance to 3.24.2 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select SEDME at any time during the contract production period and subject these SEDME to the examination specified in 4.6.1 and the tests marked "X" in Column 3 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on SEDME selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production SEDME.

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In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison SEDME to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison SEDME and shall be cause for refusal by the Government to continue acceptance of production SEDME until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on SEDME previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all SEDME accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective for major defects and 6.0 percent defective for minor defects.

Major

116. Materials and containers not as specified for Level A or B. Each incorrect material or container shall be considered one defect.

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117. Desiccant not placed in the case as specified for Level A or B.
118. Quantity of desiccant not in accordance with the referenced formula as specified for Level A or B.
119. Wood block not secured in end of box containing the tripod as specified for Level A or B.
120. Fiberboard boxes not closed and secured in accordance with the referenced method as specified for Level A or B.
121. Strapping not as specified for Level A or B.

Minor

207. Marking missing, illegible, incorrect, or incomplete for Level A, B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or B, or Commercial as specified (see 6.2).

5.1.1 Level A. All components for which cases are provided shall be placed in their respective cases and the case lids closed and secured. In addition, the case containing the instrument shall have desiccant placed within the case. The desiccant shall conform to MIL-D-3464 and shall be in the quantity required in accordance with the formula specified in MIL-P-116. The components shall be placed in the transport case as provided for and the case lid closed and secured. Components such as the tripod, for which cases are not provided, shall be placed in close-fitting boxes conforming to PPP-B-636, W5c, style optional. The box containing the tripod shall have a fitted block of wood secured in the end of the box to prevent the tripod legs from puncturing the box. Each box shall be closed and secured in accordance with Method V of the appendix to the box specification.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A.

5.1.3 Commercial. Each surveying instrument with the accessories and associated components contained in the cases provided shall be preserved to provide protection against mechanical and physical damage from the contractor to the initial destination.

5.2 Packing. Packing shall be Level A or B, or Commercial as specified (see 6.2).

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5.2.1 Level A. The components comprising a complete surveying instrument, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. The contents shall be blocked or braced within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified, (see 6.2) strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A except the box shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. The components comprising a complete surveying instrument, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, National Stock Number, quantity, contract or purchase order number and appropriate address markings on the shipping container. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. A minimum of two surveying instruments, one operating in the measuring mode and the other operating in the answering mode, are intended for the precise measurement of survey lines.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model, and number of SEDME required (see 3.2).

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(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required and number of SEDME to be furnished (see 3.3).
(e) Winterization kit is required (see 3.13).
(f) Color other than forest green (see 3.15 and 3.28).
(g) Degree of preservation and packing required (see 5.1 and 5.2).
(h) When other than Finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production SEDME from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish SEDME conforming to this specification.

6.4 Definition. The following definition shall apply throughout this specification.

6.4.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the SEDME with provisions of this specification.

6.5 Safety. The contracting officer should arrange to furnish to the contractor a copy of TB MED 270 and require the contractor to furnish certified electromagnetic radiation data of the microwave output of a representative surveying instrument of his own manufacture that complies with this specification to the US Army Environmental Hygiene Agency in accordance with AR 40-5 for the purpose of determining that all applicable safety requirements specified in 3.21 have been complied with.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each SEDME.

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6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of SEDME.

Custodians:
Army - ME
Air Force - 99

Review activities:
Army - EL, CE
DSA -GS

User activity:
Navy - MC

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-S-52156C Surveying Instrument, Distance - Measuring, Electronic Microwave

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

YES NO *(If "Yes", in what way?)*

4 REMARKS

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